

DEPARTMENT OF CSE

B.Tech — 4th Year Major Project (2025–2026)

Project Title	:	A HYBRID MACHINE LEARNING-BASED SMART HOME IOT SECURITY SYSTEM FOR THREAT DETECTION, CLASSIFICATION, AND ALERT MECHANISMS
Domain / Track	:	IOT
Student Name(s)	:	Gadivenuka Abhishek Goud - 22071A0518, Gourishetty Sai Srujith - 22071A0521, Vadaparthi Pavan Kumar – 22071A0559, Yamsani Saketh – 22071A0564
Supervisor	:	P Praveen, Assistant Professor, CSE

Problem Statement

IoT devices in smart home environments are highly vulnerable to cyberattacks due to weak security mechanisms and continuous network exposure. Existing systems often fail to effectively detect unknown attacks and accurately classify different types of threats. This creates a need for an efficient system that can monitor network traffic, detect anomalies, and classify attacks to improve IoT security.

Objectives

(1) To design and implement a real smart home IoT environment using IoT devices such as ESP32 and a smart bulb communicating over Wi-Fi. **(2)** To capture and analyse normal network traffic generated by IoT devices using monitoring tools such as Wireshark. **(3)** To simulate different network-based attacks on IoT communication and collect corresponding attack traffic data. **(4)** To preprocess and analyse both normal and attack traffic datasets for identifying behavioural patterns. **(5)** To develop and evaluate an anomaly-based intrusion detection system using machine learning models to distinguish normal and malicious IoT traffic. **(6)** To extend the system by integrating automated response and alert mechanisms for real-time IoT threat detection and mitigation.

Abstract

This project focuses on developing an IoT-based network security system for smart home environments using machine learning techniques. With the rapid growth of IoT devices, network security has become a critical concern due to the increasing number of cyber threats. The proposed system aims to detect anomalies and classify cyberattacks in IoT network traffic. A hybrid approach is used, combining Local Outlier Factor (LOF) for anomaly detection and Random Forest for multi-class attack classification. The system processes network traffic data through preprocessing steps such as data cleaning, feature encoding, scaling, and dimensionality reduction using Principal Component Analysis (PCA). The LOF model identifies deviations from normal behavior, while the Random Forest model classifies detected anomalies into specific attack types. An alert mechanism is integrated to provide real-time notifications when threats are detected. The system is expected to achieve high accuracy in anomaly detection and reasonable performance in attack classification. Overall, the project provides a practical and efficient solution for enhancing IoT network security and enabling real-time threat detection in smart home environments.

Expected Outcomes

(1) A functional IoT network security system capable of detecting anomalies and classifying attacks. (2) Trained machine learning models (LOF and Random Forest) with evaluated performance metrics. (3) An alert system that generates real-time notifications for detected threats. (4) A complete project report along with system implementation and demonstration.

Supervisor Signature